

Final Curation Log

Hyperspectral Grass Reflectance Dataset

Dataset: Pau et al. (2025) - Grass hyperspectral reflectance and traits

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Project: IS547 Data Curation, Spring 2026

Log Period: March 10 - April 21, 2026

Purpose

This log documents all curation work on the grass hyperspectral dataset from Dryad. I'm following the CURATE(D) framework and keeping track of what I checked, what issues I found, and decisions I made along the way.

Week 1: Check & Understand (March 10-16)

March 10 - Downloaded and verified files

Downloaded the complete dataset from Dryad. Got 3 files:

- Pau_et_al_2025_spectra.csv (45.2 MB)
- Pau_et_al_2025_traits.csv (64.8 KB)
- README.txt (2.1 KB)

Checked that all files opened without errors. Files look complete - no obvious corruption.

March 11 - File structure examination

Spectra file: 426 rows × 2,151 columns. First column is sample_id, then 2,150 wavelength columns named X400.0 through X2400.0. These must be reflectance at different wavelengths (0.5 nm intervals).

Traits file: 426 rows × 23 columns. Has sample_id plus taxonomic info, site codes, and various trait measurements.

Both files have exactly 426 rows - suggests they should match up 1:1.

March 12 - Verified how files link

Checked if sample_id column exists in both files - yes! Tested in R to make sure all IDs match:

```
all(spectra$sample_id %in% traits$sample_id) # TRUE
```

Perfect 1:1 match - all 426 samples in spectra file have corresponding trait records. No orphaned records.

Decision: sample_id is the definitive join key between files.

March 13 - README review

Original README is pretty brief (~500 words). It has basic info like authors, DOI, funding, but missing a lot of details:

- No variable definitions or units
- Doesn't explain how files link together
- No info about site locations
- Missing data collection details

Started drafting an improved README with all the missing sections.

Week 2: Explore & Evaluate (March 17-23)

March 17 - Missing value analysis

Calculated % missing for each trait variable. Key findings:

- Taxonomic data: 0% missing (complete)
- Morphological traits: 8-18% missing
- Chemical traits: 14-20% missing

Interesting: nitrogen_content and carbon_content have EXACTLY the same samples missing (61 samples). Makes sense - they're probably measured together on a CN analyzer.

Overall ~10-20% missing trait data. Not ideal but workable.

March 18-19 - Value range checks

Checked if trait values fall in reasonable ranges. Most look good:

- SLA: 8.45-45.89 mm²/mg (reasonable for grasses)
- Nitrogen: 0.89-4.73% (normal range)
- Reflectance: 0.003-0.895 (all values between 0-1, good)

Potential issues: Found 3 samples with very high SLA (>40) and 2 with unusually high chlorophyll values. Could be real or could be errors - flagged for authors to verify.

March 20 - Units problem

Big issue: units are NOT specified for most trait variables. The README doesn't say and column names don't include units. Had to infer from value ranges and literature:

- SLA: probably mm²/mg based on values
- LDMC: probably mg/g
- Chlorophyll: totally unclear - could be µmol/m² or SPAD units

This is a major reusability barrier. Added to priority recommendations - authors need to specify units explicitly.

March 21 - Site codes

Dataset has 11 different site codes (KNZ, CDR, SEV, SGS, HYS, etc.) but they're not defined anywhere in the deposit. Based on the paper and my knowledge of LTER sites:

- KNZ = Konza Prairie, Kansas
- CDR = Cedar Creek, Minnesota
- SEV = Sevilleta, New Mexico

But no coordinates or detailed site descriptions in the deposit. Added site info to the enhanced README.

Week 3: Document & Test (March 24-30)

March 24-26 - Created documentation

Finished the enhanced README and data dictionary:

- **README:** Expanded from 500 to ~1,500 words with clear sections on files, variables, sites, methods, and examples
- **Data dictionary:** Defined all 2,174 variables (2,151 spectral + 23 traits) with types, units, ranges, and missing patterns

Tried to be realistic about what I could infer vs. what needs author clarification. Used tables to make it easier to scan.

March 27-28 - Basic reproducibility test

Tried to reproduce a simple analysis from the paper - comparing C3 vs C4 grass spectra.

Success: Was able to merge files, extract VIS/NIR spectra, calculate means by photosynthetic pathway, and plot results. The patterns match what's in the paper - C3 and C4 grasses have different spectral signatures.

Blocked: Couldn't do a proper site-level comparison because site codes aren't well documented. Would need site coordinates and descriptions to interpret site effects properly.

Documented the R code in the README as an example for users.

March 29 - Quality summary

Overall assessment:

- **Good:** Files are well-structured, join cleanly, values are plausible, >99% spectral data complete
- **Needs improvement:** Missing units, limited site info, no preprocessing documentation, 10-20% missing trait data
- **Priority fixes:** Add units for all variables, define site codes, document spectral processing steps

Week 4: Synthesize & Recommend (March 31 - April 6)

April 1-2 - Finalized evaluation report

Completed the curation evaluation report by adding the remaining sections:

- Reproducibility testing results: Documented Part 1 success (lineage analysis) and Part 2 blocking issue (site analysis)
- Metadata enhancement recommendations: Proposed specific keywords, geographic coverage terms, and temporal scope improvements for Dryad deposit
- Complete author clarification list: Finalized 12 priority questions organized into HIGH (4 questions), MEDIUM (5 questions), and LOW (3 questions) categories
- Implementation guidance: Provided step-by-step instructions for how authors could incorporate enhanced documentation
- Preservation assessment: Evaluated CSV format suitability for long-term archiving (excellent), file organization (clear), and documentation completeness (needs improvement)

The final report is now comprehensive and provides clear, actionable recommendations.

April 3-4 - Completed reproducibility documentation

Finished the reproducibility check report by adding:

- Detailed code appendix: Complete R script with step-by-step comments showing exactly what I ran for both analyses
- Quantitative comparison to paper: Matched my C3 mean reflectance values to Figure 2 in the paper - values align within expected measurement error
- Additional analyses attempted: Documented my attempt to correlate SLA with NIR reflectance (partially successful but limited by missing SLA values)

Overall conclusion: 60% reproducible - strong enough for basic lineage comparisons, insufficient for site-level or trait-correlation analyses.

April 5-6 - Metadata recommendations

Developed specific Dryad metadata improvements:

- **Keywords to add:** "hyperspectral remote sensing", "C3 C4 photosynthesis", "grassland ecology", "plant functional traits", "LTER"
- **Geographic coverage:** Added bounding box coordinates covering all 11 sites (approximately 32°N-47°N, 94°W-109°W)
- **Temporal coverage:** Based on paper methods, samples collected 2018-2020 growing seasons

These changes would significantly improve dataset discoverability in searches for spectral ecology and grassland research.

Week 5: Package & Refine (April 7-13)

April 7-8 - Final quality checks

Reviewed all deliverables for completeness and consistency:

- Enhanced README: Verified all cross-references work, code examples run correctly, site information accurate
- Data Dictionary: Double-checked all 2,174 variable definitions, confirmed missing percentages match my calculations
- Evaluation Report: Ensured all findings are supported by evidence in the log, recommendations are realistic
- Reproducibility Check: Verified R code runs from scratch, environmental details complete

Fixed minor inconsistencies (e.g., site code KNZ vs Konza, wavelength ranges) across all documents.

April 9-10 - GitHub repository organization

Organized all artifacts in GitHub repository (<https://github.com/AshishGole2000/Data-Curation-Project>):

- Created clear folder structure: /deliverables, /documentation, /code
- Added comprehensive README.md explaining repository contents and how to use artifacts
- Uploaded all final versions: Enhanced README, Data Dictionary (PDF + MD), Curation Log, Evaluation Report, Reproducibility Check
- Included R code examples and sample output files

Repository is now well-organized and accessible for peer review and instructor evaluation.

April 11-13 - Peer review preparation

Prepared artifacts for peer review process:

- Created ARTIFACTS_README.md explaining package contents and how reviewers should evaluate each deliverable
- Added explicit version markers ("FINAL - April 2026") to all documents
- Ensured all file formats are accessible (Word for editing feedback, PDF for stable viewing, Markdown for web display)

All deliverables are now ready for peer and instructor review.

Week 6: Final Report & Submission (April 14-21)

April 14-16 - Final narrative report

Wrote the comprehensive final project report following the submission rubric:

- Explained scenario, use case, and curation goals with clear motivation
- Thoroughly characterized dataset including source, provenance, format, constraints
- Addressed ethical, legal, and policy compliance (NSF data sharing, CC0 license, no human subjects)
- Documented complete curation workflow following CURATE(D) framework

- Detailed metadata and documentation improvements with examples
- Explained provenance and data lineage tracking methods
- Demonstrated reproducibility and transparency through test results
- Justified archiving approach (Dryad repository, CSV format, open license)

Report length: ~2,100 words. Includes all APA-formatted references.

April 17-18 - Final log polish

Completed and polished this curation log:

- Added all Week 4-6 activities to bring log to 100% completion
- Ensured chronological flow and decision rationale is clear throughout
- Cross-referenced log entries with final deliverables for consistency
- Added final summary section documenting full project scope

This log now provides complete provenance for all curation decisions and activities.

April 19-20 - Submission preparation

Prepared final submission package:

- Converted final narrative report to PDF
- Verified GitHub repository is public and accessible
- Created supplementary materials ZIP archive as backup
- Confirmed all deliverables meet rubric requirements
- Double-checked APA citation format for all references

April 21 - Project submission

Submitted final project:

- Final report PDF uploaded
- GitHub repository link provided: <https://github.com/AshishGole2000/Data-Curation-Project>
- All supplementary materials accessible

Project Summary

Total duration: 6 weeks (March 10 - April 21, 2026)

Framework: CURATE(D) - Check, Understand, Request, Augment, Transform, Evaluate, Document

Key Accomplishments:

- Verified dataset integrity: 426 samples, 2,174 variables, 100% join integrity
- Identified 4 major reusability barriers with priority levels
- Created enhanced README (3x improvement: 500→1,500 words)
- Developed comprehensive data dictionary (all 2,174 variables defined)

- Tested reproducibility: 60% successful (lineage analysis works, site analysis blocked)
- Produced 6 ranked recommendations for authors
- Compiled 12 priority clarification questions
- Enhanced Dryad metadata (keywords, geographic/temporal coverage)

Impact:

This curation work makes the dataset significantly more reusable for independent researchers. The enhanced documentation addresses the most critical barriers (missing units, undefined sites, unclear file relationships) while maintaining transparency about what was inferred vs. what needs author clarification. An independent researcher can now confidently reproduce basic lineage-based analyses and understands exactly what additional information would enable more complex analyses.

Lessons Learned:

- Even well-structured data needs explicit documentation - file quality ≠ reusability
- Authors often don't realize what context external users need (e.g., site definitions obvious to LTER researchers)
- Reproducibility testing is essential - reveals documentation gaps that aren't obvious from inspection alone
- Prioritization matters - not all issues have equal impact on reusability
- Transparency is key - clearly documenting what I inferred vs. what I know is crucial for trust